

CIS 106 Introduction to Programming
Using the Basic If Statement

1. Allow a user to enter the quantity of an item. If the quantity is greater than or equal to 1000, the unit price should be \$3.00. For quantities under 1000 the unit price is \$5.00. Compute extended price to be quantity x unit price. Compute tax to be 7% of the extended price. The total is computed as extended price plus the tax. Display the quantity, unit price, extended price, tax and total.

INPUT	PROCESS	OUTPUT
Enter the quantity of an item	If the quantity is greater than or equal to 1000, the unit price is \$3.00. Compute the extended price as quantity x unit price. Compute the tax as 7% of this extended price. The total cost is the extended price plus tax.	Display the quantity, unit price, extended price, tax, and total.
	If the quantity is less than 1000, the unit price is \$5.00. Compute the extended price as quantity x unit price. Compute the tax as 7% of this extended price. The total cost is the extended price plus tax.	

2. The program asks the user for an item and quantity. Determine the unit price of the item based on the chart below. Compute the extended price to be quantity x unit price. Display the item, unit price and extended price. Note: if the item entered is not A then assume the item is B. No need to check for B.

Item	Unit Price
A	\$10.00
B	\$20.00

(Note: assume the user will enter the data correctly. Assume if they enter capital A then \$10.00 gets assigned to unit price variable. Any other entry is assumed to be a capital B whether they enter B or not. Therefore, you only need a relational condition for A. This makes the if statement easier and removes data validation from the program which could get quite complex).

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if item == "A":
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    Unit_price = 10.00
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else:
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    Unit_price = 20.00
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INPUT	PROCESS	OUTPUT
Item and quantity	Check if the item is "A". If Item = "A": Unit_price is = 10.00. Else: Unit_price is = 20.00. Compute the extended price to be quantity x unit price.	Display the item, unit price, and extended price.

3. Enter the number of books to order and cost per book. If the order total is over \$50.00 shipping is free. If the order total is \$50.00 or under charge \$25 shipping. Display the order total and shipping charge (note 0 should display for a free shipping charge).

INPUT	PROCESS	OUTPUT
Number of books	If the order total is greater than \$50.00, shipping is free.	Display the order total and shipping charge (shipping charge is \$0 if order is greater than \$50.00)
Cost per book	If the order is less than \$50.00, add a \$25.00 shipping charge.	

4. The warrantee of an appliance depends on the cost of the appliance. For appliances over \$1,000 the warrantee cost is 10% of the price. For appliances \$1,000 or less the warrantee cost is 5% of the price. The user will enter the name and cost of an appliance. Display name and cost of appliance, the cost of the warrantee and the total (cost of the appliance + warranty).

INPUT	PROCESS	OUTPUT
Enter the name and cost of an appliance.	If the appliance costs more than \$1000, the warranty cost is 10% of the appliance value. Compute the total cost of the appliance using the cost of the appliance + the cost of the warranty.	Display name and cost of appliance, the cost of the warranty, and the total (cost of the appliance + warranty).
	If the appliance is less than or equal to \$1000, the	

INPUT	PROCESS	OUTPUT
	warranty cost is 5% of the appliance value. Compute the total cost of the appliance using the cost of the appliance + the cost of the warranty.	

5. Enter the user's last name, number of dependents and gross income. Compute adjusted gross income to be gross income minus dependents times \$12000. Next determine an income tax rate. Adjusted gross incomes over \$50,000 have a tax rate of 20%. Adjusted gross incomes \$50,000 or under have a tax rate of 10%.

Once you determine the tax rate, compute income tax to be adjusted gross income times tax rate. If the income tax is less than 0, set the income tax to \$100.

Display last name, gross income, number of dependents, adjusted gross income, and income tax.

INPUT	PROCESS	OUTPUT
Enter the user's last name, number of dependents, and gross income.	Compute adjusted gross income to be gross income- [# of dependents x \$12000].	Display last name, gross income, number of dependents, adjusted gross income, and income tax.
	Determine an income tax rate using the adjusted gross income over \$50,000. If adjusted gross income is greater than \$50,000, the tax rate is 20%. If adjusted gross income is less than or equal to \$50,000, the tax rate is 10%.	
	Compute income tax as adjusted gross income x tax rate. If income tax is less than 0, set the income tax to \$100.	

**Example Programs - You do not have to do these problems.
So I did not.**